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THESIS OUTLINE REPORT

**SYMBOL DETECTION AND
FINE-GRAINED CLASSIFICATION IN
P&ID**

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1 Introduction

1.1 General Introduction

Piping and Instrumentation Diagrams (P&ID) are the classic engineering documents used to specify process equipment, pipelines, valves, instrumentation, and control loops in industries such as oil and gas, chemical processing, and power generation. They encode this information through a large, standardized vocabulary of graphical symbols, and the automated interpretation of these symbols from raster images is a long-standing computer-vision problem. Among the many sub-problems that P&IDs raise — symbol detection, text recognition, line tracing, and graph extraction — **symbol detection and classification** is the most fundamental: it produces the localized, labeled symbol instances that every other interpretation task depends on, and it remains the principal bottleneck in published pipelines, where fine-grained errors among visually similar valve and instrument classes are the main remaining source of error.

In recent years, deep learning techniques have achieved significant progress in object detection and information extraction from engineering drawings. A growing body of research has applied convolutional and transformer-based detectors to P&ID symbol recognition, reporting strong precision and recall on standard benchmarks. However, these reported results have been obtained almost exclusively on synthetic datasets covering a narrow symbol vocabulary — typically around 32 classes — and conforming to a single drawing standard. More fundamentally, the existing literature reports aggregate detection scores but rarely isolates **fine-grained classification performance** among visually similar symbol families such as valve variants, instrument bubbles, and fittings. This is exactly where engineering errors are most costly. Whether current approaches transfer to the conditions encountered in industrial practice — scanned legacy drawings, dense and overlapping symbol layouts, broader symbol vocabularies, and the long tail of safety-critical rare classes — therefore remains a question that aggregate mAP cannot answer.

Symbol detection and classification in P&IDs is challenging primarily because of **inter-class visual similarity**. A comprehensive P&ID may contain 50 to 100 distinct symbol categories, and many of these categories belong to *visually similar groups* that share a common base geometry and differ only in small internal markings: the eight or more valve variants (gate, globe, ball, butterfly, needle, plug, pinch, diaphragm) are the typical example, with instrument bubbles and pipe fittings exhibiting analogous within-group ambiguity. Distinguishing among classes within such a group is a *fine-grained classification* problem, distinct from the coarse localization problem solved well by modern detectors. Secondary challenges — symbol density and occlusion, scan-induced image degradation, and company-specific drawing variations — affect localization quality and dataset construction respectively, and are addressed at pre-processing time; they are well-studied in the broader detection literature and are not the focus of this study.

These observations expose a specific weakness in current P&ID detection pipelines that has not been systematically addressed: while dense object detection on engineering drawings is increasingly well-studied, the **fine-grained classification step within detection pipelines has received limited attention** in the P&ID literature, despite being the dominant source of error on visually similar classes. Standard detection heads couple coarse localization and fine classification into a single softmax over all K classes, devoting only a fraction of model capacity to the within-group discrimination that matters most for engineering correctness. Compounding this, comprehensive symbol vocabularies of 50 or more classes exhibit a severe long-tail frequency distribution in which rare but safety-critical symbols are consistently under-represented in training data — a challenge that is most serious at the fine-grained classification stage, since the model must learn slight within-group distinctions from very few examples. Aggregate detection metrics hide both of these failure modes. Secondary issues such as high-resolution tiling and synthetic-to-real generalization remain relevant and are addressed as supporting architectural decisions, but they are not the central technical problem of this work.

1.2 Thesis Objectives and Tasks

Motivated by the considerations above, this study proposes a modular **detection-then-fine-grained-classification pipeline** for P&IDs, in which a dedicated fine-grained classification stage handles within-group discrimination separately from the detector. The objectives of the thesis and the concrete tasks

required to deliver them are summarized below; the methodology and implementation chapters that follow develop each item in detail.

1.2.1 Thesis Objectives

This thesis aims to deliver the following:

- **Detection-plus-FGC pipeline:** a pipeline that separates coarse symbol detection (localization plus super-category) from fine-grained within-group classification, in contrast to the single-softmax design of standard one-stage detectors.
- **Fine-grained classification stage:** a stage operating on cropped symbol regions, designed to address the visually similar groups (valve family, instrument-bubble family, fitting family) that dominate the residual error in prior P&ID detection work.
- **FGC evaluation protocol:** within-group confusion matrices, per-family accuracy, top- k classification accuracy, and rare-class macro F1, reported alongside standard detection metrics to expose error modes that aggregate mAP conceals.
- **Dataset strategy:** a combination of existing public synthetic data, programmatic synthesis with both coarse and fine labels, and a small real-world held-out evaluation set for measuring the synthetic-to-real generalization gap.

1.2.2 Thesis Tasks

The objectives above are realized through the following concrete tasks:

1. Assemble and unify the training data: reuse existing public synthetic datasets and implement a **synthetic-generation pipeline** that emits both coarse and fine labels per symbol.
2. Collect and manually annotate a small set of real-world P&IDs to serve as a **held-out evaluation set**.
3. Build the detection pipeline: implement the preprocessing and **overlap-preserving tiling** stage for high-resolution inputs, train a one-stage detector for **localization and coarse classification**, and combine per-patch detections into a coherent **diagram-level symbol inventory**.
4. Implement and train the **fine-grained classification stage** on cropped symbol regions, with per-class balancing for rare fine classes.
5. Define and implement the evaluation protocol, including **detection metrics** and **fine-grained classification metrics**, with per-class and per-family reporting.

2 Background & Problem Analysis

2.1 Piping and Instrumentation Diagrams

2.1.1 Standardization of Symbol Systems

Piping and Instrumentation Diagrams are governed by globally recognized standards to ensure consistency across different engineering fields and organizations. The foundational standard is **ISO 10628**, which specifies general principles for constructing flow diagrams in the process industry, including the classification and symbolic representation of equipment, piping, and measuring elements. This standard organizes symbols into thematic groups, each focusing on a specific category of industrial equipment or control element. This grouping structure helps engineers quickly identify symbols and ensures consistent interpretation in multinational projects. Besides **ISO 10628-2**, the **ISA -5.1** standard issued by the International Society of Automation is also widely used in practice, particularly in the field of industrial automation. This standard specifies symbols and identifiers for measuring devices, control loops, and signal types, forming a closely complementary symbol system to **ISO 10628-2**.

Within the scope of this thesis, **ISO 10628-2** is chosen as the symbol vocabulary set C that the system needs to detect and classify, due to its widespread use and application in practice. Although **ISA-5.1** is also a viable option, it is not publicly available and requires a license, making it unsuitable for the open research purposes of this thesis.

2.1.2 ISO 10628-2 Symbol Sheets

- Sheet 1 – Vessels and Tanks; Columns with Internals

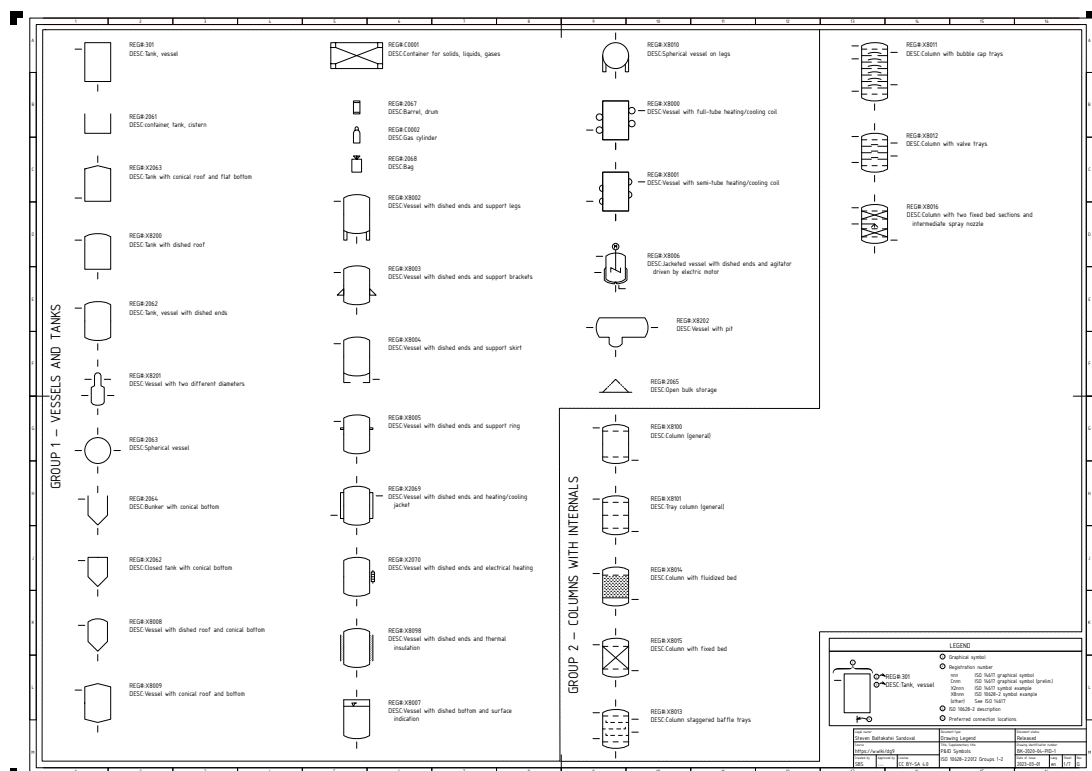


Figure 1: ISO 10628-2 Sheet 1

- Sheet 2 – Heat Exchangers, Steam Generators, Furnaces and Recooling Devices

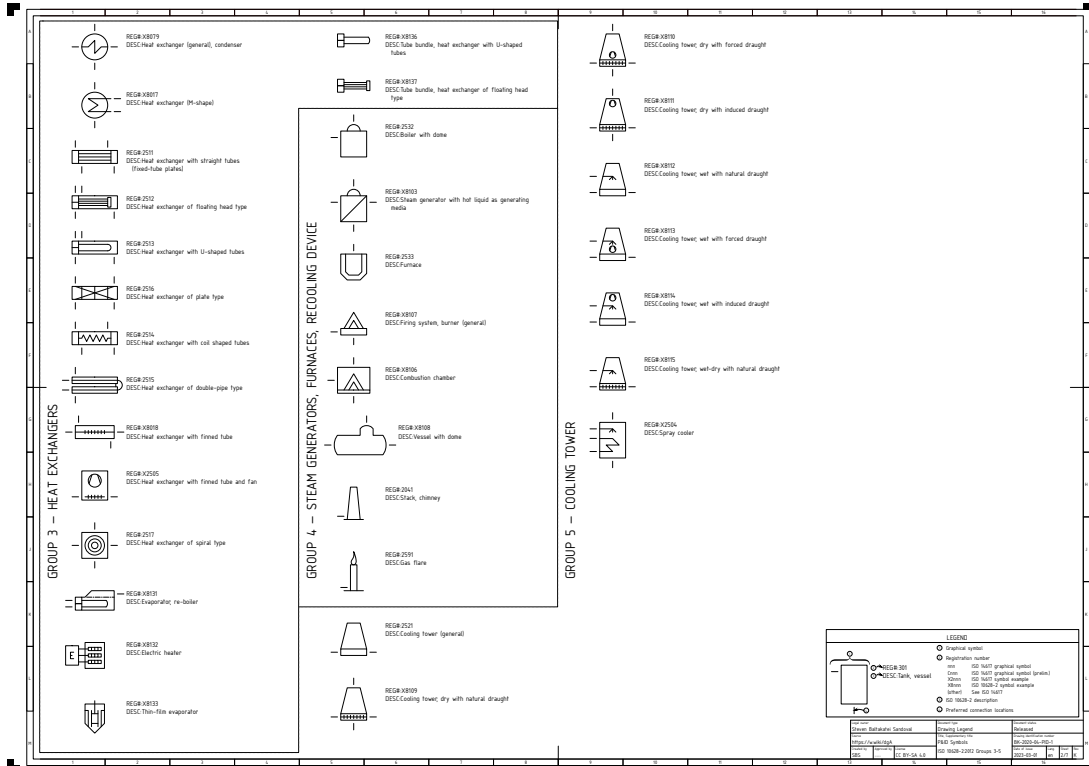


Figure 2: ISO 10628-2 Sheet 2

• Sheet 3 – Filters, Screening Devices, Separators and Centrifuges

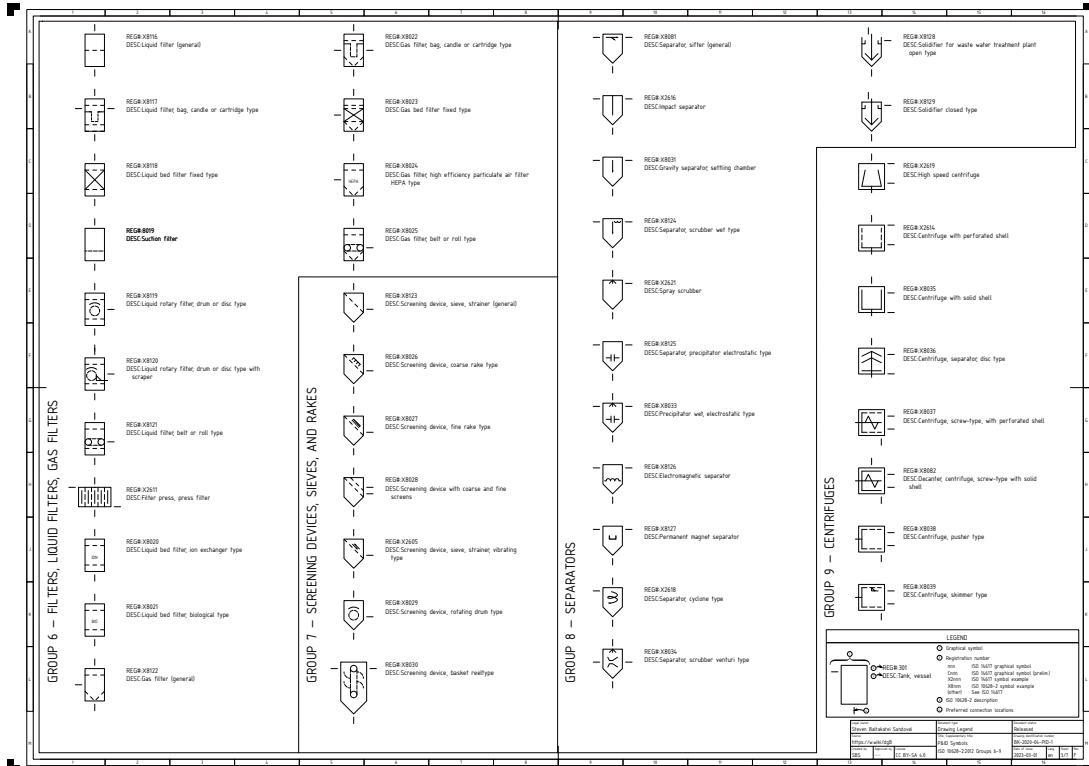


Figure 3: ISO 10628-2 Sheet 3

• Sheet 4 – Dryers, Crushers, Mixers, Pumps, Compressors and Blowers

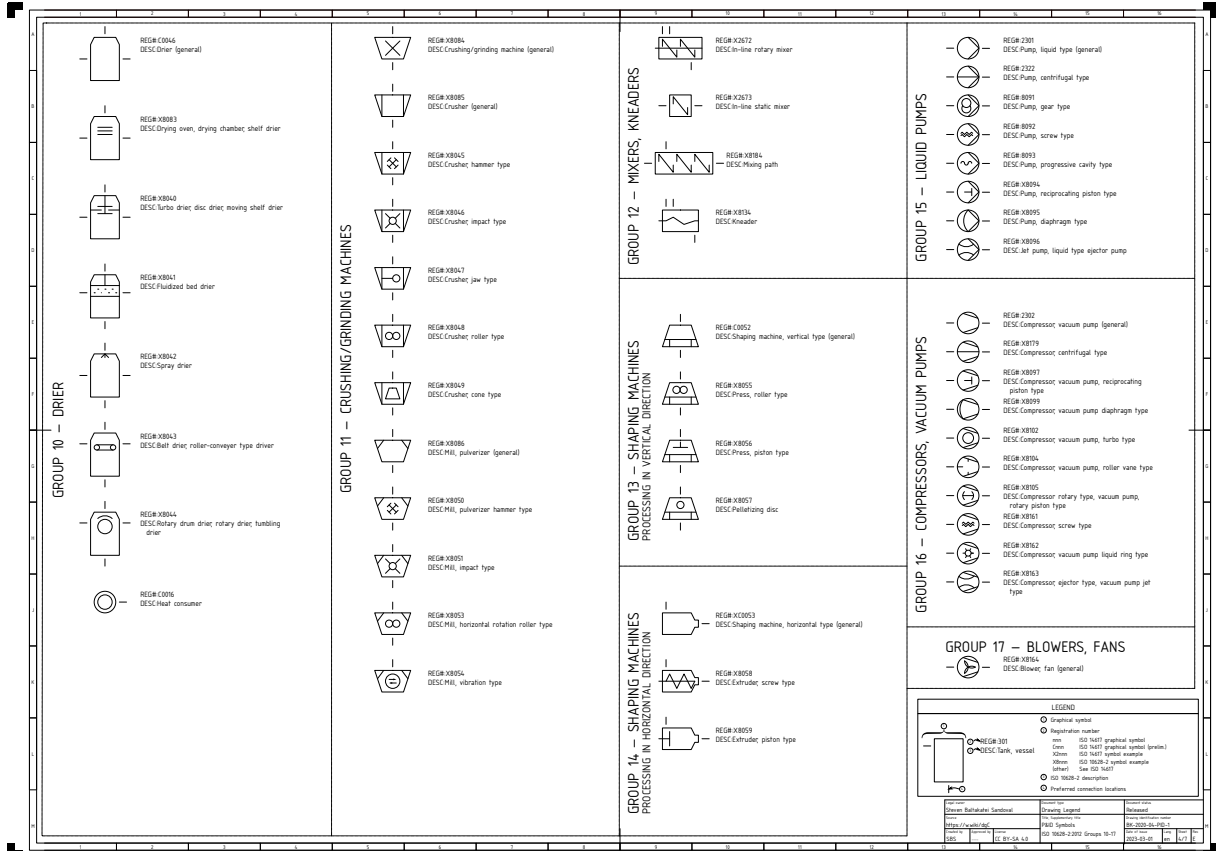


Figure 4: ISO 10628-2 Sheet 4

• Sheet 5 – Lifting & Transport Equipment, Proportioners and Engines

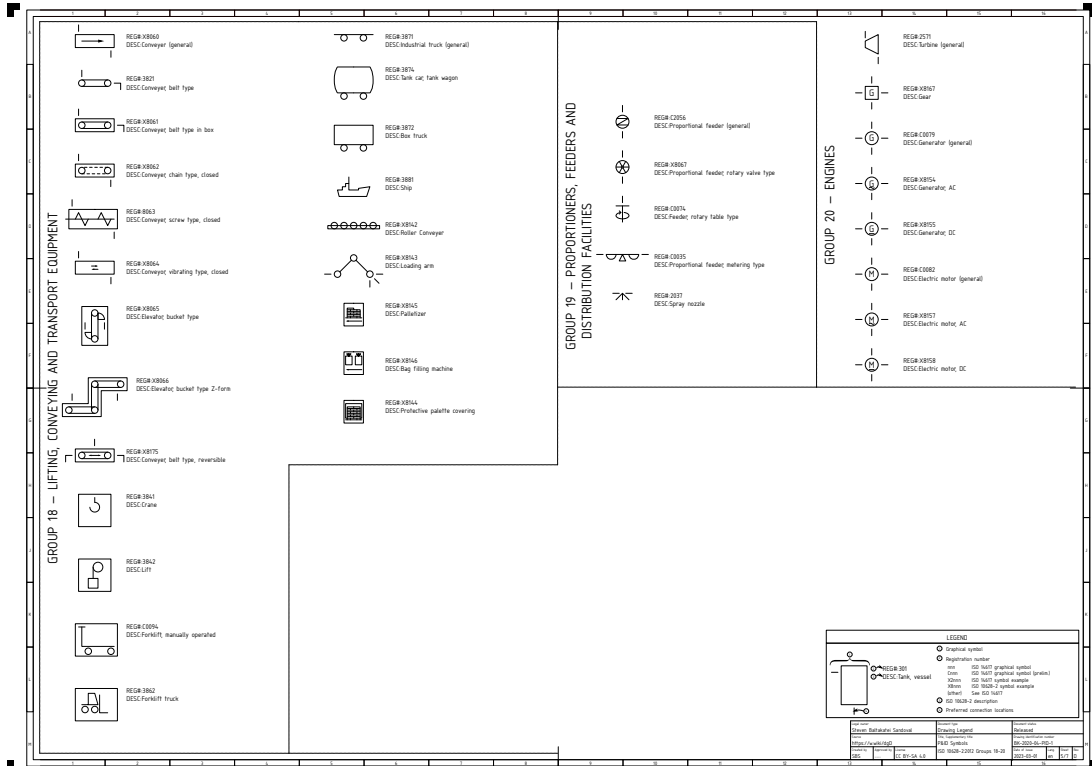


Figure 5: ISO 10628-2 Sheet 5

• Sheet 6 – Valves, Fittings and Piping Symbols

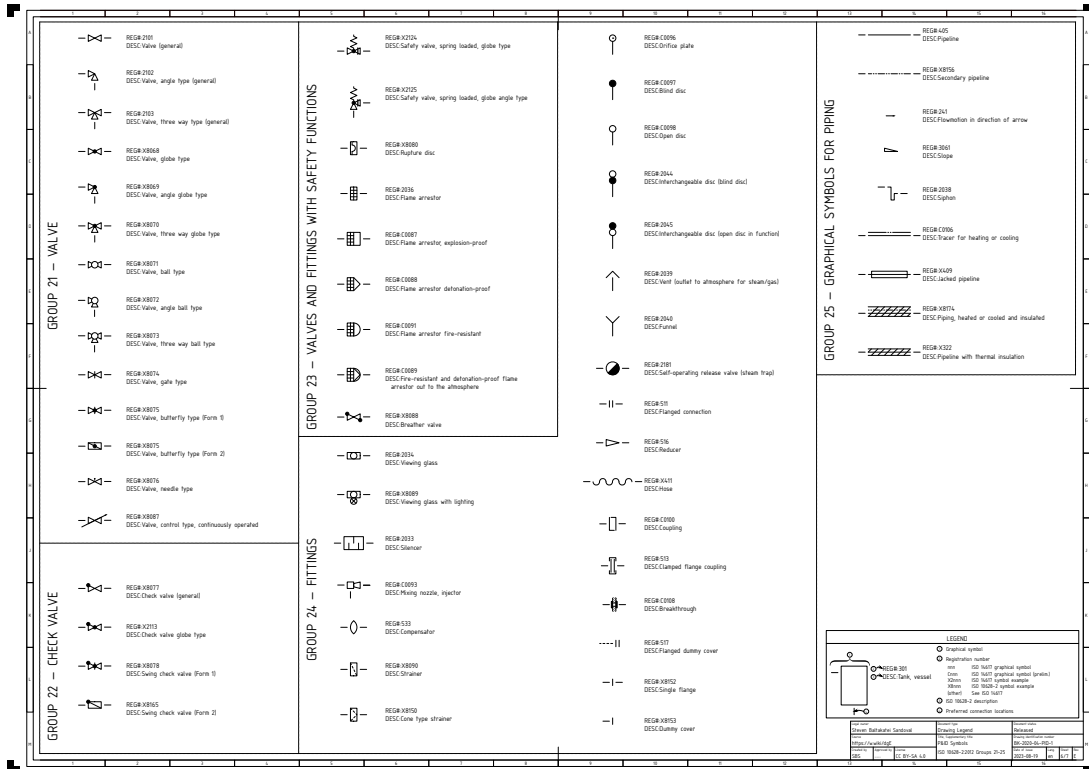


Figure 6: ISO 10628-2 Sheet 6

• Sheet 7 – Apparatus Elements, Internals, Agitators and Built-in Components

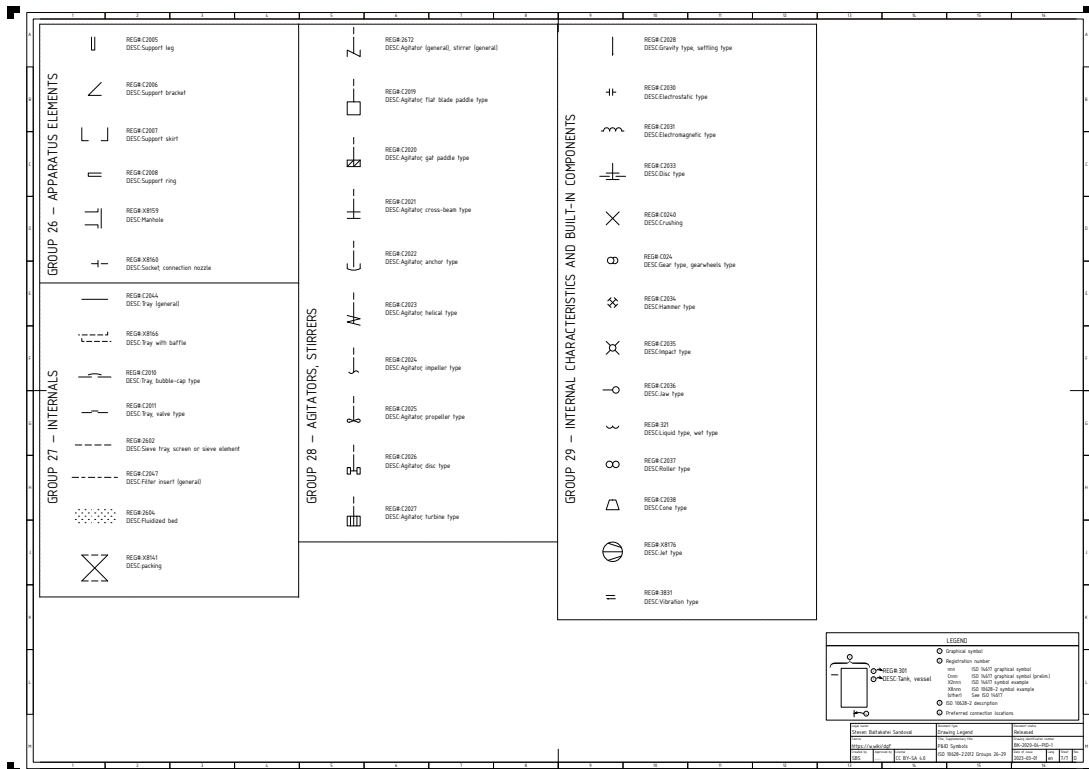


Figure 7: ISO 10628-2 Sheet 7

2.2 Symbol Detection and Fine-Grained Classification

2.2.1 Overview

The core task addressed in this study is **symbol detection and fine-grained classification in P&IDs**. We treat this as two coupled but technically distinct sub-problems:

- **Sub-problem 1 — Detection (localization and coarse classification).** Given a rasterized diagram image, produce a set of bounding boxes that tightly enclose each symbol instance and assign each box to a *super-category* (e.g., **valve**, **instrument_bubble**, **pump**, **fitting**). This sub-problem is concerned primarily with *where* symbols are and *what kind* they broadly belong to.
- **Sub-problem 2 — Fine-Grained Classification (FGC).** Given a localized symbol region, assign it to a *fine class* within its super-category, where the fine vocabulary contains visually similar groups (e.g., the valve family {gate, globe, ball, butterfly, needle, plug, pinch, diaphragm}, the instrument-bubble family, the fitting family). This sub-problem is concerned with *within-group discrimination*, where class boundaries are defined by small internal markings rather than overall geometry.

The full task can be formalized as follows. Given a rasterized P&ID image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote the height, width, and number of channels respectively, the goal is to produce a structured set of detections:

$$\mathcal{D} = \{(b_i, c_i, s_i) \mid i = 1, 2, \dots, N\} \quad (1)$$

where each detection tuple consists of:

- $b_i = (x_i, y_i, w_i, h_i)$ — the bounding box of the i -th detected symbol, defined by its center coordinates and dimensions in pixel space;
- $c_i \in \mathcal{C}$ — the predicted symbol class, drawn from a predefined fine vocabulary $\mathcal{C} = \{c^{(1)}, c^{(2)}, \dots, c^{(K)}\}$ of K symbol categories. The vocabulary is hierarchically organized into a small set of super-categories (associated with Sub-problem 1) and the fine classes within each super-category (associated with Sub-problem 2);
- $s_i \in [0, 1]$ — the confidence score associated with the detection.

This formulation positions the overall problem as an **object detection** task augmented with a **fine-grained classification** stage. A key design principle of this work is that the two sub-problems are evaluated separately: localization quality is reported through standard detection metrics, while fine-grained classification quality is reported through within-group accuracy and confusion analyses on cropped symbol regions. This separation is itself one of the contributions of the study, as it exposes error modes that aggregate detection metrics conceal.

The input to the system is a rasterized image of a P&ID page, typically obtained through one of the following pipelines:

- **Scanned documents:** Physical drawings digitized via flatbed or large-format scanners, commonly stored as TIFF or JPEG files at resolutions between 150 and 400 DPI.
- **PDF rasterization:** Vector-based PDF files converted to raster images using rendering engines, potentially losing sub-pixel precision during conversion.
- **Native digital exports:** CAD-generated drawings exported directly as high-resolution PNG or TIFF files, offering the highest geometric fidelity.

Due to the large physical dimensions of engineering drawings (often A0 or A1 format), high-resolution scans can produce images exceeding $10,000 \times 7,000$ pixels. As a result, preprocessing steps such as **patch-based cropping** or **multi-scale tiling** are typically required before inference to accommodate the fixed input dimensions of deep learning models.

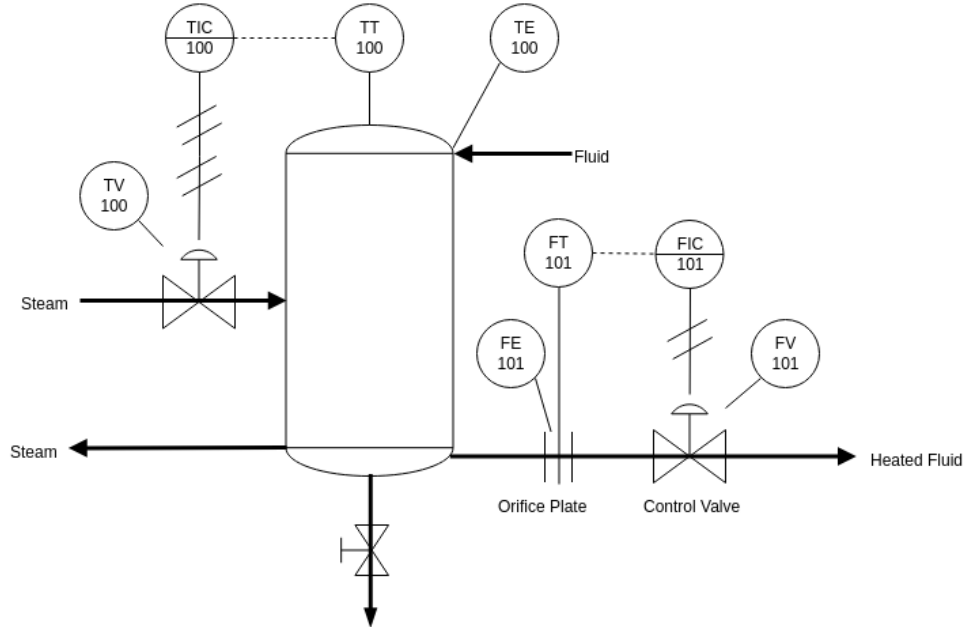


Figure 8: An example of *P&ID* is used as input

The expected output of the system is a structured annotation of all symbol instances present in the diagram. For each detected symbol, the output encodes:

- **Spatial localization:** A bounding box b_i defining the pixel-level extent of the symbol within the diagram image.
- **Semantic classification:** A fine class label c_i identifying the engineering type of the detected symbol (e.g., `centrifugal_pump`, `gate_valve`, `pressure_transmitter`), together with its associated super-category.
- **Detection confidence:** A scalar score s_i reflecting the model's certainty, used for threshold-based filtering and ranking.

The output is intended for direct use in symbol-inventory tasks — for example, counting safety-critical instruments, auditing symbol vocabulary coverage, or producing a labeled symbol manifest for a diagram set — and as a self-contained deliverable. Downstream digitization tasks such as pipeline-symbol association, connectivity graph construction, and integration into digital engineering platforms are outside the scope of this study and are not modeled by the output schema.

2.2.2 Key Challenges

The central technical challenge of this study is *within-group discrimination among visually similar symbol classes* — the fine-grained classification problem. The remaining challenges (density and occlusion, image quality, cross-company variation) affect localization quality and dataset construction respectively; they are well-studied in the broader detection literature and are addressed in this work through preprocessing and dataset design rather than treated as research contributions. The challenges are presented in order of centrality to this study.

Inter-class Visual Similarity and Visually Similar Groups (Central Challenge)

A comprehensive P&ID may contain upward of 50 to 100 distinct symbol categories. Critically, these categories are not uniformly distributed across the visual space: they cluster into *visually similar groups* whose members share a common base geometry and differ only in small internal markings. The most prominent examples are:

- **The valve family** — gate, globe, ball, butterfly, needle, plug, pinch, and diaphragm valves are all variations of the same bow-tie / hourglass outline, distinguished by an internal stem, disc, or fill pattern. This family alone contains eight or more fine classes that share an essentially identical bounding-box silhouette.
- **The instrument-bubble family** — pressure, temperature, flow, and level instruments (transmitters, indicators, controllers, recorders) are all represented as circles, distinguished only by alphanumeric tags inside the circle. The visual geometry is identical across the entire family; the semantic class is encoded in text.
- **The fitting family** — reducers, couplings, flanges, blind discs, and orifice plates share short, line-aligned silhouettes that are easily confused with one another and with short pipeline segments.

This structure makes within-group discrimination a **fine-grained classification problem** in the technical sense established in the broader vision literature: the discriminating features are local, fine in scale, and small compared to the whole object as a whole. Standard object detectors — which use a single softmax over the full class vocabulary $\mathcal{C}$ — devote only a fraction of their classification capacity to within-group separation, because their classification head must simultaneously distinguish coarse super-categories (where features are global and the problem is easy) and fine within-group classes (where features are local and the problem is hard). In practice, this coupling produces detectors with high coarse-category accuracy but elevated confusion within visually similar groups, and these within-group confusions are the dominant source of residual error reported in prior P&ID detection works.

The central technical question of this study follows directly: *can a dedicated fine-grained classification stage, operating on cropped symbol regions and trained with FGC-aware data and metrics, significantly reduce within-group confusion compared to a single-pass detector?* The methodology in Chapter 4 is structured around this question.

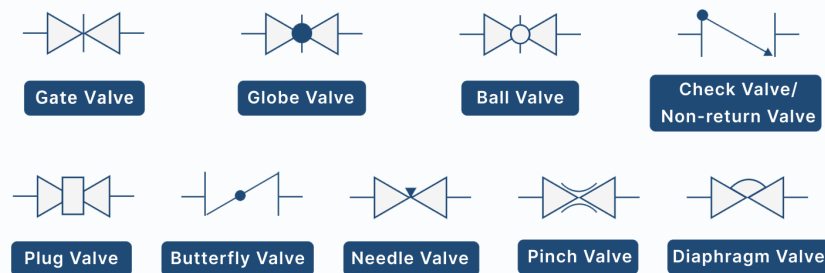


Figure 9: The valve family as a representative visually similar group. The eight illustrated valve types share an essentially identical outer silhouette and are distinguished only by small internal markings (stem, disc, fill pattern), making them a canonical fine-grained classification problem within the P&ID vocabulary.

Symbol Density, Overlap, and Occlusion (Localization Challenge)

P&IDs are inherently information-dense documents. A single sheet may contain several hundred symbols distributed across a constrained drawing area, with adjacent symbols, crossing pipelines, and annotation text frequently overlapping symbol boundaries. This density is primarily a *localization* challenge — it affects where bounding boxes can be reliably placed — and is addressed in this study through the preprocessing and tiling choices (overlap-preserving SAHI tiling). It is not a focus of the FGC contribution, and is mentioned here for completeness rather than as an open research problem this study aims to resolve.

Image Quality Degradation and Noise (Robustness Consideration)

A large proportion of P&ID documents in industrial practice exist as scanned copies of hand-drawn or legacy printed drawings, introducing noise, uneven illumination, ink bleed, skew, and compression artifacts. These artifacts degrade symbol geometry and complicate downstream detection. In this work they are treated as a robustness consideration handled at the preprocessing stage (contrast normalization, denoising, skew correction) rather than as a central technical challenge.

Company-Specific and Non-Standard Symbol Variations (Scope Boundary)

Although ISO 10628 and ISA 5.1 define internationally recognized symbol conventions, adherence to these standards varies considerably across organizations: many engineering companies, EPC contractors, and national industries employ proprietary symbol libraries that deviate from or extend the standard set. **This study does not claim cross-company generalization.** The symbol vocabulary $\mathcal{C}$ is fixed to the ISO 10628-2 / ISA 5.1 conventions as represented in the training data; the system is evaluated on diagrams matching that vocabulary. Cross-standard generalization and domain adaptation across proprietary symbol libraries are acknowledged as important open problems but are explicitly out of scope.

Class Imbalance and Long-Tail Distribution (Fine-Grained Classification Problem)

A comprehensive industrial P&ID may reference upward of 50 to 100 distinct symbol categories, but their frequency within any given diagram is highly non-uniform. A small set of common symbols — instrument bubbles, gate valves, pipe fittings, and process lines — accounts for the majority of all symbol instances, while a long tail of less frequent but operationally critical symbols (safety relief valves, emergency shutoff valves, specialized control devices) may appear only a handful of times across an entire dataset, or be absent from individual diagrams entirely.

This long-tail distribution is most consequential *at the fine-grained classification stage*, not the detection stage. Coarse super-category statistics tend to be more balanced — valves and instrument bubbles as a whole are not rare — but the fine classes within those super-categories are severely imbalanced. The model must learn the subtle within-group distinctions for rare fine classes (e.g., needle valve, pinch valve, safety relief valve) from very few training examples, while the gradient signal from common fine classes (e.g., gate valve, pressure indicator) dominates training. Compounding this, the rare fine classes are frequently the most safety-critical, so aggregate metrics such as mean Average Precision can mask poor performance precisely where reliability matters most.

Addressing this imbalance is therefore primarily an FGC design problem. It motivates the FGC-specific design choices in Section 4.2.3 — in particular, training the FGC head on cropped symbol patches (where aggressive class balancing and targeted augmentation are practical), loss reweighting with Focal Loss [1] for rare classes, and per-class FGC-specific evaluation reporting (Section 2.2.3).

2.2.3 Evaluation Metrics

The evaluation protocol of this study reports two complementary families of metrics: **detection metrics** that quantify localization quality and aggregate classification accuracy, and **fine-grained classification metrics** that isolate within-group discrimination performance. Reporting both families together — rather than relying on aggregate mAP alone — is one of the contributions of this work.

Detection Metrics: For the detection sub-problem, we adopt evaluation metrics standard to object detection tasks, adapted to the specific requirements of the P&ID domain.

(a) Intersection over Union (IoU): The spatial accuracy of a predicted bounding box $\hat{b}$ relative to a ground-truth box b^* is measured by the Intersection over Union:

$$\text{IoU}(\hat{b}, b^*) = \frac{|\hat{b} \cap b^*|}{|\hat{b} \cup b^*|} \quad (2)$$

A detection is considered a *true positive* if $\text{IoU} \geq \theta$, where θ is a predefined threshold. The standard threshold $\theta = 0.5$ is commonly adopted; however, in P&ID contexts where symbols are densely packed, stricter thresholds (e.g., $\theta = 0.75$) may be appropriate to penalize imprecise localization.

(b) Precision and Recall: At a fixed IoU threshold θ and confidence threshold τ , standard detection precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

where TP , FP , and FN denote the number of true positives, false positives, and false negatives respectively. In the P&ID domain, **high recall** is particularly critical, as missed symbols (false negatives) represent gaps in the engineering record that could compromise downstream analysis or safety validation.

(c) Average Precision (AP) and mean Average Precision (mAP): Average Precision summarizes the precision-recall trade-off for a single class by computing the area under the precision-recall curve:

$$\text{AP} = \int_0^1 p(r) dr \approx \sum_k [r_k - r_{k-1}] p_k \quad (4)$$

The mean Average Precision (mAP) aggregates AP across all K symbol classes:

$$\text{mAP} = \frac{1}{K} \sum_{k=1}^K \text{AP}_k \quad (5)$$

Two standard variants are commonly reported: mAP_{50} (evaluated at $\text{IoU} = 0.50$) and $\text{mAP}_{50:95}$ (averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05), the latter being the primary metric in the COCO benchmark [2].

(d) Per-class AP: Given the large number of symbol categories and their imbalanced frequency in real P&IDs, reporting only aggregate mAP may obscure poor performance on infrequent symbol classes. Therefore, per-class AP is reported alongside aggregate metrics. Per-class AP captures rare-class detection quality but, as discussed below, it does not in itself isolate *within-group* confusion — which is the specific failure mode that FGC is designed to address. For that purpose, the dedicated FGC metrics introduced next are required.

Fine-Grained Classification Metrics: The detection metrics above quantify how well symbols are localized and, in aggregate, how well they are labeled. They do not, however, expose the structure of classification errors: an mAP figure of 0.85 is consistent with both a model that confuses random pairs of classes uniformly and a model whose errors are concentrated entirely within the valve family — two very different failure modes with very different operational consequences. Because within-group confusion is the dominant residual error mode for P&ID symbol classification, and because reducing it is the central technical goal of this work, we report a family of **fine-grained classification metrics** that target it directly.

(a) Within-Group Confusion Matrices: For each visually similar group $G \subset \mathcal{C}$ defined in Section 2.2.2 (the valve family, instrument-bubble family, fitting family, etc.), we report a restricted confusion matrix $M^G \in \mathbb{N}^{|G| \times |G|}$ over predictions whose ground-truth class lies in G and whose predicted class also lies in G . The diagonal of M^G corresponds to correct within-group classification; the off-diagonal mass quantifies within-group confusion. Restricting attention to G isolates the FGC problem cleanly: it factors out coarse-category errors (which are dominated by detection quality) and exposes the within-group failure pattern that the FGC module is designed to correct.

(b) **Within-Group Accuracy:** The scalar summary of M^G is the within-group accuracy:

$$\text{Acc}_G = \frac{\sum_{c \in G} M_{cc}^G}{\sum_{c \in G} \sum_{c' \in G} M_{cc'}^G} \quad (6)$$

We report Acc_G separately for each defined group G (e.g., $\text{Acc}_{\text{valve}}$, $\text{Acc}_{\text{bubble}}$, $\text{Acc}_{\text{fitting}}$). These per-family scores are the primary headline metric for the FGC contribution.

		Predicted class				
		Gate	Globe	Ball	Butterfly	Needle
True class	Gate	142	18	5	1	0
	Globe	22	118	7	2	1
	Ball	4	6	89	3	0
	Butterfly	2	3	4	54	1
	Needle	0	2	1	2	17

Correct within-group
 Within-group confusion
 Rare / zero

Figure 10: Example within-group confusion matrix for the valve family. Each row is the true class; each column is the predicted class. The green diagonal counts correct classifications; the red off-diagonal counts within-group confusions (e.g., a gate valve predicted as a globe valve). Aggregate detection metrics such as mAP collapse this entire 5×5 structure into a single number, hiding where the errors are concentrated — which is exactly what the FGC-specific metrics in this section are designed to expose. The within-group accuracy for this example is $\text{Acc}_G \approx 0.84$. Counts are illustrative.

(c) **Top- k Accuracy on the FGC Head:** The fine-grained classification module operates on cropped symbol patches and produces a probability distribution over fine classes. Top- k accuracy on this distribution, evaluated on a held-out set of ground-truth crops, characterizes the FGC head independently of detection quality. We report Top-1 and Top-3 accuracy. Top-3 in particular reveals whether the FGC head is ranking the correct class highly even when it is not the argmax — a useful diagnostic for ambiguous cases where the head may legitimately be uncertain.

(d) **Macro-Averaged F1 on Rare Classes:** To surface long-tail FGC quality independently of aggregate mAP, we report macro-averaged F1 restricted to the set of rare fine classes $\mathcal{C}_{\text{rare}} \subset \mathcal{C}$ (defined as classes appearing in fewer than a fixed threshold of training instances — a typical choice is the bottom decile by training frequency):

$$\text{F1}_{\text{macro,rare}} = \frac{1}{|\mathcal{C}_{\text{rare}}|} \sum_{c \in \mathcal{C}_{\text{rare}}} \text{F1}_c \quad (7)$$

Because macro-averaging weights each rare class equally regardless of support, this metric is sensitive to failure on *any* rare class — including the safety-critical ones that aggregate mAP would otherwise drown out.

These FGC-specific metrics are reported alongside the standard detection metrics in all experiments, and constitute one of the contributions of this work: they expose the fine-grained classification quality that aggregate mAP hides, and they enable direct measurement of the marginal contribution of the FGC module in the ablation experiments planned in Chapter 5.

Table 1: Summary of evaluation metrics for P&ID symbol detection and fine-grained classification. The protocol reports detection metrics (top group) and fine-grained classification metrics (bottom group) together; the latter are a contribution of this work.

Metric	Notation	Role
<i>Detection metrics (localization + aggregate classification)</i>		
Intersection over Union	IoU	Spatial localization accuracy of predicted bounding boxes
Precision	P	Fraction of detections that are correct; penalizes false positives
Recall	R	Fraction of ground-truth symbols detected; penalizes missed symbols
Average Precision	AP	Area under precision–recall curve for a single class
Mean Average Precision	mAP ₅₀	Average AP across all classes at IoU threshold 0.50
Mean Average Precision	mAP _{50:95}	Average AP across all classes and IoU thresholds 0.50–0.95
Per-class AP	AP _k	Class-level breakdown for identifying weak classes
<i>Fine-grained classification metrics (within-group discrimination)</i>		
Within-group confusion	M^G	$ G \times G $ matrix isolating within-group errors for group G
Within-group accuracy	Acc _G	Per-family accuracy (valve family, bubble family, fitting family)
Top- k accuracy	Top- k	FGC-head accuracy on cropped patches ($k = 1, 3$)
Rare-class macro F1	F1 _{macro,rare}	Long-tail FGC quality independent of aggregate mAP

3 Related Work

3.1 Existing Dataset

The availability of large-scale, high-quality annotated datasets is a prerequisite for training and benchmarking deep learning models. In the P&ID domain, however, dataset construction faces a set of structural challenges that fundamentally distinguish it from mainstream computer vision tasks. This section surveys the datasets currently available for P&ID symbol detection and classification research, analyzes their characteristics and limitations, and discusses strategies that have been proposed to address the persistent data scarcity problem.

3.1.1 Dataset Construction

Two factors make P&ID dataset construction uniquely difficult. First, **industrial confidentiality**: real-world P&ID documents encode sensitive information about plant topology, equipment specifications, and process design. As a result, operating companies are generally unwilling to share diagrams publicly, and academic researchers rarely have access to large collections of authentic engineering drawings. Second, **annotation cost**: labeling a single high-resolution P&ID sheet — which may contain several hundred symbol instances across 50 or more categories — requires significant time investment and domain expertise, as correct identification of symbols often demands engineering knowledge beyond what a general annotator possesses.

These structural barriers have produced a research landscape characterized by small, domain-specific, and predominantly synthetic datasets, with very limited standardized benchmarking infrastructure.

3.1.2 Synthetic Datasets

Given the scarcity of publicly available real-world P&ID data, the dominant strategy in the literature has been the programmatic generation of synthetic diagrams. Synthetic datasets are constructed by algorithmically placing standard symbol templates onto randomly generated pipeline networks, optionally augmented with noise, rotation, scale variation, and scan simulation to improve domain realism[3].

The most widely adopted public dataset in this category is **Dataset-P&ID**, introduced alongside the Digitize-PID framework [4]. Dataset-P&ID comprises 500 synthetically generated P&ID sheets annotated with bounding boxes for 32 symbol classes, covering complex symbols that are structurally similar to one another as well as simpler basic shapes. The dataset was among the first publicly released resources for P&ID recognition research and has since been adopted as a training and evaluation basis in multiple subsequent works [5, 6, 7]. An extended variant of this dataset, comprising 500 P&ID sheets with 32 symbol categories and incorporating different types of noise, was later used by the Azure Digitization project, demonstrating precision and recall performance exceeding 90% under controlled conditions.

A smaller community-contributed dataset is available through **Roboflow Universe** [8], providing a limited collection of annotated P&ID images suitable for prototyping but insufficient in scale for robust model training. Similarly, **Kaggle** hosts a P&ID symbol dataset [9] intended for symbol extraction tasks, though both platforms offer datasets that are narrow in symbol vocabulary and diagram diversity.

3.1.3 Real-world Datasets

Authenticated real-world P&ID datasets are exceedingly rare in the published literature. Paliwal et al. evaluated their Digitize-PID system on a private real-world dataset of 12 anonymized P&ID sheets, noting that despite the small sheet count, each diagram contains hundreds of visual and textual components — making the evaluation substantively meaningful.

More recently, Stürmer et al. introduced PID2Graph, described as the first publicly available P&ID dataset containing real-world diagrams with full graph-level annotations — encompassing both symbol

bounding boxes and their interconnection edges. **For the purposes of this study, only the bounding-box and symbol-label component of PID2Graph is relevant.** The graph-level (connectivity) annotation is outside the scope of the symbol detection and fine-grained classification task addressed here, and is treated as a complementary contribution of that dataset rather than a benchmark dimension this study evaluates against. The dataset’s scale and class coverage on the symbol-detection axis alone remain limited by the constraints of real-world data acquisition.

3.1.4 Limitations of Current Datasets

Table 2 summarizes the principal datasets available for P&ID symbol detection and classification research.

Table 2: Summary of publicly available datasets for P&ID symbol detection and classification. Graph-level annotations (where present) are out of scope for this study and are noted only for completeness.

Dataset	Source	Diagrams	Classes	Type	Annotation	Public
Dataset-P&ID [4]	TCS Research	500	32	Synthetic	Bounding box + lines	Yes
PID2Graph [6]	Academic	Small	–	Real-world	Full graph (symbols + edges)	Yes
Roboflow P&ID [8]	Community	<100	Limited	Mixed	Bounding box	Yes
Kaggle P&ID [9]	Community	Limited	Limited	Mixed	Bounding box	Yes
Private datasets	Industrial	Varies	Varies	Real-world	Varies	No

The datasets currently available exhibit four systemic limitations that constrain the development of generalizable P&ID symbol detection and classification models:

(1) Limited scale. Even the largest public dataset (Dataset-P&ID) contains only 500 synthetic diagrams — a quantity substantially smaller than the tens of thousands of images typically used to train robust object detectors in natural image domains.

(2) Narrow symbol vocabulary. Most datasets cover between 7 and 50 symbol classes, whereas a comprehensive industrial P&ID may reference upward of 100 distinct symbol types. Models trained on narrow vocabularies cannot be deployed in industrial settings without significant retraining, and the narrow vocabulary in particular obscures within-group fine-grained classification quality, since visually similar groups (valve family, instrument-bubble family) are often only partially represented.

(3) Synthetic-to-real domain gap. Synthetic diagrams, while useful for bootstrapping model training, differ systematically from authentic engineering drawings in line weight variation, symbol rendering style, spatial layout patterns, and annotation conventions. Models trained exclusively on synthetic data frequently exhibit degraded performance on real-world inputs.

(4) Single-standard coverage. Existing datasets are typically derived from a single symbol standard (most commonly a variant of ISO or ISA conventions), providing no coverage of the cross-company and cross-standard variability observed in industrial practice.

These limitations collectively highlight the need for either larger and more diverse real-world dataset collection efforts, or improved synthetic data generation strategies capable of faithfully modeling the visual and structural characteristics of authentic P&ID documents.

3.2 Traditional Approaches

The earliest efforts in automated P&ID analysis relied on traditional computer vision and pattern recognition techniques. These methods approach symbol detection and classification through hand-crafted feature representations and explicit geometric reasoning, without requiring large annotated datasets. **This section reviews approaches to the symbol detection and classification sub-problem only;** traditional methods for line tracing, junction analysis, and graph extraction — which are central to end-to-end P&ID digitization but outside the scope of this study — are not discussed. The remainder of the

section surveys the principal traditional paradigms, evaluates their respective strengths and limitations, and synthesizes the fundamental challenges they face when applied to real-world P&ID diagrams.

3.2.1 Template Matching

Template matching is the most straightforward approach to symbol recognition, operating by sliding a reference symbol image — the template across the query diagram and computing a pixel-level similarity measure at each spatial position. Common similarity functions include Normalized Cross-Correlation (NCC) and the Sum of Absolute Differences (SAD). A detection is triggered at positions where the similarity score exceeds a predefined threshold.

$$\text{NCC}(x, y) = \frac{\sum_{u,v} [I(x+u, y+v) - \bar{I}] [T(u, v) - \bar{T}]}{\sqrt{\sum_{u,v} [I(x+u, y+v) - \bar{I}]^2 \cdot \sum_{u,v} [T(u, v) - \bar{T}]^2}} \quad (8)$$

where I is the diagram image, T is the template, and $\bar{I}$, $\bar{T}$ denote their respective local mean intensities.

Early applications of template matching to P&ID symbol detection demonstrated viability under controlled conditions — particularly for clean, digitally generated diagrams where symbols closely match standard templates. However, the method carries fundamental limitations. It is inherently sensitive to geometric transformations: any deviation in scale, rotation, or aspect ratio between the template and the target instance degrades matching accuracy. More critically, the method requires a separate template for each symbol class and orientation variant, making it impractical for vocabularies of 50 to 100 symbol categories. Partial occlusion from overlapping symbols or crossing pipeline lines further reduces reliability.

3.2.2 Feature-based Matching

To address the sensitivity of template matching to geometric transformations, feature-based methods replace dense pixel comparison with sparse local descriptors. Algorithms such as SIFT [10], SURF [11], and ORB [12] extract keypoints and compute invariant descriptors at important image locations, enabling robust matching under scale and rotation changes. Symbol recognition is then performed by matching descriptor sets between a query region and a library of reference symbols.

Feature-based approaches offer improved robustness compared to template matching and require no model training. However, their applicability to P&IDs is significantly constrained by the nature of engineering line drawings. Unlike natural scene images, P&ID symbols are composed primarily of thin strokes and geometric primitives with minimal texture, resulting in sparse and unstable keypoint distributions. Under such conditions, SIFT and SURF detectors yield few reliable keypoints per symbol, and false matches are frequent — particularly among instrument bubbles and valve symbols whose structural repetition produces ambiguous descriptor sets. Furthermore, matching complexity scales with the size of the template library, making real-time inference across large symbol vocabularies computationally expensive.

3.2.3 Shape Descriptor Methods

Shape descriptor methods depart from local feature matching by representing each symbol as a compact global or semi-global geometric signature. A classifier — typically a Support Vector Machine (SVM) or k -Nearest Neighbour (k -NN) — is then trained to map these descriptors to symbol class labels. Commonly used descriptors in P&ID-related work include:

- **Hu Moments** [13]: a set of seven moment invariants derived from image intensity distributions, providing invariance to translation, scale, and rotation.
- **Zernike Moments**: orthogonal complex moments computed over a unit disk, offering greater discriminative power and noise robustness than Hu Moments.

- **Fourier Descriptors:** frequency-domain representations of the symbol contour, encoding shape periodicity in a compact and rotation-invariant form.
- **Shape Context** [14]: a histogram-based descriptor that captures the spatial distribution of contour points relative to each other, enabling flexible shape comparison via the Hungarian algorithm.

These approaches are computationally efficient and perform well under clean, isolated symbol conditions. However, global shape descriptors are inherently sensitive to contour distortion caused by image noise, scan artifacts, or partial occlusion — all of which alter the boundary geometry that the descriptor encodes. Moreover, representing a symbol as a single global vector discards structural information about the arrangement of internal sub-components, limiting discriminability among visually similar symbol classes — precisely the within-group fine-grained discrimination on which this study focuses.

3.2.4 Limitations for P&ID

While each of the above paradigms has demonstrated utility under restricted conditions, their collective performance degrades substantially when confronted with the full complexity of real-world P&IDs. Table 3 provides a comparative assessment across four critical dimensions.

Table 3: Comparative assessment of traditional symbol recognition methods across key criteria relevant to P&ID applications.

Method	Noise Robustness	Overlap Handling	Scalability	Cross-standard
Template Matching	Low	Poor	Poor	Poor
Feature-based	Moderate	Poor	Moderate	Moderate
Shape Descriptor	Low–Moderate	Poor	Good	Moderate

Beyond these individual weaknesses, three systemic limitations characterize the entire class of traditional methods when applied to P&ID symbol detection and classification:

(1) Poor robustness under domain shift. All traditional approaches rely on hand-crafted representations that are suited to the visual vocabulary of a specific symbol set or standard. When applied to diagrams produced under a different company convention or regional standard — as is common in industrial practice — these representations fail to generalize without manual redesign of templates, descriptors, or graph grammars. There is no mechanism for automatic adaptation to unseen symbol variants.

(2) Absence of contextual reasoning. Traditional methods evaluate each candidate region in isolation, without regard for the surrounding visual context. In P&IDs, however, the identity of a symbol is often clarified by its spatial relationship to neighboring elements: the type of pipeline connected to it, the equipment it is associated with, or its positional role in the process flow. This contextual information — which is easily used by human engineers — is entirely discarded by existing traditional approaches.

(3) Inability to scale to real-world diagram complexity. Industrial P&IDs routinely contain several hundred symbols per sheet, with high spatial density, frequent symbol overlap, and pipeline lines that traverse and intersect the drawing plane. None of the methods surveyed above have demonstrated robust end-to-end performance under these conditions without substantial manual preprocessing or post-correction. The reliance on clean, isolated symbol instances as a requirement for reliable recognition is incompatible with industrial deployment requirements.

These limitations collectively motivate the adoption of deep learning-based approaches, which — as reviewed in the following section — offer the capacity to learn discriminative feature representations directly from data, handle visual variability through large-scale training, and leverage spatial context through convolutional and attention-based architectures.

3.3 Deep Learning Approaches

The limitations of traditional methods have motivated a substantial body of research applying deep learning to symbol detection and classification in engineering diagrams. Unlike hand-crafted pipelines, deep learning models learn hierarchical feature representations directly from annotated data, enabling greater robustness to visual variability, noise, and domain-specific complexity. This section surveys the principal deep learning paradigms applied to object detection and segmentation, evaluates their architectural characteristics, and assesses their suitability for the P&ID detection and fine-grained classification task. At the end of each architecture subsection, a brief *FGC angle* note identifies what each architecture does — or fails to do — with respect to within-group fine-grained classification, the central technical concern of this study.

3.3.1 General Framework for Detection

Modern object detectors share a common three-stage architectural blueprint, illustrated in Figure 11:

- **Backbone:** A deep convolutional network (e.g., ResNet [15], VGG [16]) or Vision Transformer that extracts hierarchical feature representations from the input image at multiple spatial scales.
- **Neck:** A feature aggregation module — most commonly a Feature Pyramid Network (FPN) [17] or Path Aggregation Network (PANet) — that fuses multi-scale feature maps to enrich both semantic and spatial resolution, enabling detection of objects at varying sizes.
- **Detection Head:** The final prediction module, which maps aggregated features to bounding box coordinates and class probability distributions. The architectural design of the head is the primary axis along which detector families diverge, and — as discussed below — the head is also the point where fine-grained classification capacity is determined.

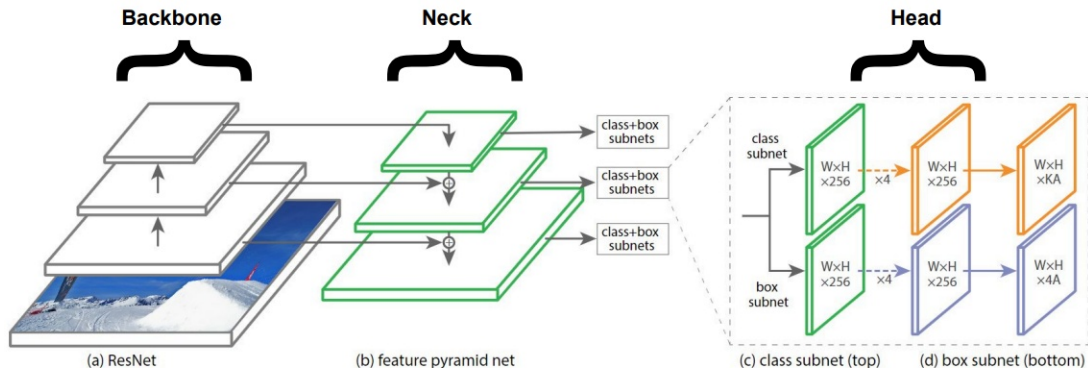


Figure 11: General architectural pipeline of a modern deep learning object detector, consisting of a backbone for feature extraction, a neck for multi-scale feature aggregation, and a detection head for bounding box regression and classification.

The shift from hand-crafted to learned representations is particularly significant in the P&ID domain, where the feature space required to distinguish visually similar symbol classes — differing only in subtle internal geometry — is difficult to specify manually but can in principle be learned from sufficient annotated examples.

3.3.2 Two-stage Detectors

Two-stage detectors decompose the detection problem into two sequential sub-tasks: region proposal generation followed by per-region classification and bounding box refinement. The pioneering work of Girshick et al. introduced R-CNN [18], which applied a CNN to warped image regions proposed by selective search. Subsequent works — Fast R-CNN [19] and Faster R-CNN [20] — progressively unified

the pipeline, leading to the Region Proposal Network (RPN): a lightweight convolutional network that shares features with the detection backbone to generate candidate regions end-to-end. Mask R-CNN [21] further extended this framework by adding a parallel branch for pixel-level instance segmentation.

The two-stage paradigm offers high detection accuracy, particularly for small and densely packed objects, owing to the dedicated refinement pass applied to each candidate region. This property is advantageous in P&IDs, where symbols are small relative to the full diagram resolution and may be spatially clustered. Several studies have adopted Faster R-CNN as the detection backbone for P&ID symbol recognition [22] [23], reporting competitive localization accuracy on standard symbol vocabularies.

However, the sequential architecture introduces computational overhead that limits throughput on high-resolution diagram images. Furthermore, the tiling strategies required to process large-format P&ID sheets — often exceeding $10,000 \times 7,000$ pixels — can fragment symbols across patch boundaries, degrading recall unless explicit overlap and stitching mechanisms are employed.

FGC angle. The per-region classification head in two-stage detectors operates on pooled features from each proposal, which in principle gives it dedicated capacity for fine-grained discrimination. In practice, however, the head is still a single softmax over the full vocabulary $\mathcal{C}$, and its features are inherited from a backbone optimized jointly for localization and classification; there is no architectural separation between coarse super-category and fine within-group decisions, and reported P&ID studies do not isolate within-group accuracy.

3.3.3 One-stage Detectors

One-stage detectors eliminate the region proposal stage by predicting bounding boxes and class scores directly from dense feature map locations in a single forward pass. SSD [24] and the YOLO family [25] established this paradigm, trading a modest accuracy reduction for substantially improved inference speed.

The YOLO architecture has undergone extensive iterative refinement. YOLOv3 [26] introduced multi-scale prediction heads to improve small object detection. YOLOv5 and subsequent versions incorporated advanced data augmentation, improved anchor assignment strategies, and optimized training pipelines. YOLOv8 [27] and the more recent YOLOv11 adopt anchor-free detection heads and decoupled classification-regression branches, achieving accuracy competitive with two-stage detectors while retaining high throughput. RetinaNet [1] addressed the class imbalance problem inherent to dense prediction through the Focal Loss, significantly improving performance on hard negative examples.

In the P&ID research literature, one-stage detectors — particularly YOLOv5 and YOLOv8 — have emerged as the most widely adopted architecture [28, 29, 30], owing to their favorable balance between accuracy and deployment efficiency. Their anchor-based (and increasingly anchor-free) designs can be adapted to the characteristic size distribution of P&ID symbols through domain-specific configuration. Nevertheless, earlier YOLO variants exhibited reduced sensitivity to small and densely overlapping symbols, a limitation that has been progressively addressed in recent releases through improved multi-scale feature fusion and loss formulations.

FGC angle. Standard YOLO heads use a single softmax over all K classes at every dense feature location, which couples coarse localization and fine classification capacity into one shared parameter set. As K grows to include the full valve and instrument-bubble families, the within-group decision boundary must be learned from a small slice of total head capacity, and is practically the dominant residual error mode. This observation motivates the detection-then-FGC cascade design adopted in this study (Section 4.2.3), in which a dedicated classifier on cropped symbol regions takes over within-group discrimination.

3.3.4 Transformer-based Detectors

The introduction of the Detection Transformer (DETR) [31] marked a fundamental departure from convolutional detection paradigms. DETR reformulates object detection as a direct set prediction problem: a CNN backbone extracts image features, which are encoded by a Transformer encoder and decoded by a fixed set of learned query embeddings through a Transformer decoder. Each query predicts one

object, and the model is trained end-to-end using a bipartite matching loss that assigns predictions to ground-truth objects without requiring non-maximum suppression (NMS) or anchor design.

The self-attention mechanism in the Transformer encoder allows DETR to model long-range spatial dependencies across the entire image in a single layer, enabling global contextual reasoning that convolutional architectures can only approximate through deep stacking. This property is particularly relevant for P&IDs, where the identity of a symbol may be identified by its spatial relationship to distant equipment or pipeline routes.

Deformable DETR [32] addressed the slow convergence and high computational cost of the original DETR by replacing dense attention with sparse deformable attention, attending only to a small set of informative sampling points per query. This modification also enables multi-scale feature processing, improving detection of small symbols. RT-DETR [33] further advanced the framework toward real-time inference while preserving end-to-end training.

Recently, transformer-based architectures have begun to be explored in the P&ID domain. Notably, Stürmer et al. [6] proposed a Relationformer-based approach that jointly extracts symbols and their interconnections from P&IDs, reporting over 25% improvement in edge detection accuracy over modular baselines. Nevertheless, the adoption of transformer models in this domain remains limited compared to the YOLO family, and large-scale benchmarking across diverse industrial datasets is still lacking — representing an active and promising research direction.

FGC angle. Transformer detectors share the single-softmax classification head of their convolutional counterparts and therefore inherit the same coupling between coarse and fine classification capacity. Their architectural advantage — long-range attention — could in principle help disambiguate visually similar symbols through context (e.g., the type of pipeline connected to a valve), but no published P&ID work has yet isolated and evaluated this effect specifically on within-group fine-grained discrimination.

3.3.5 Segmentation-based Approaches

Beyond bounding box detection, segmentation-based methods produce pixel-level outlines of symbol regions, offering finer spatial localization. Two principal formulations are relevant to P&ID analysis:

Semantic segmentation assigns a class label to every pixel in the diagram, enabling discrimination between symbol regions, pipeline line regions, text annotations, and background. Fully Convolutional Networks (FCN) [34] and encoder-decoder architectures such as U-Net [35] are commonly employed for this task. In the P&ID context, semantic segmentation has been applied as a preprocessing step to separate symbol regions from the dense network of pipeline lines before symbol-level classification is performed [36, 37, 38].

Instance segmentation, exemplified by Mask R-CNN [21], extends two-stage detection by predicting a binary pixel mask for each detected object instance alongside its bounding box and class label. This formulation can distinguish individual symbol instances even when they are in close proximity, providing more precise spatial boundaries than axis-aligned bounding boxes.

Segmentation approaches offer advantages in scenarios where symbol boundaries are irregular or where precise delineation is required for downstream graph construction. However, their adoption as primary recognition methods in P&ID pipelines has been limited by the substantially higher annotation cost: pixel-level mask labeling is significantly more labor-intensive than bounding box annotation, worsening the data scarcity challenges already prevalent in this domain. Practically, segmentation is most commonly employed as a complementary module — for line-symbol separation or region refinement — rather than as a standalone recognition system.

FGC angle. Segmentation provides pixel-level localization but uses the same per-pixel or per-instance softmax classifier as detection architectures; it offers no inherent advantage for within-group fine-grained classification, and its higher annotation cost makes the long-tail fine-class data problem worse rather than better.

3.3.6 Comparative Summary

Table 4 combines the evaluation of the four deep learning paradigms surveyed above across dimensions relevant to P&ID symbol detection and fine-grained classification. While one-stage detectors — particularly the YOLO family — currently represent the dominant approach in applied P&ID research owing to their practical efficiency, Transformer-based architectures present an underexplored research frontier with structural advantages well-suited to the contextual complexity of engineering diagrams. Critically, none of the surveyed paradigms architecturally separates coarse and fine classification capacity, which leaves within-group fine-grained discrimination as a shared open weakness across the field.

Table 4: *Comparative assessment of deep learning detection paradigms with respect to P&ID symbol detection and fine-grained classification requirements.*

Paradigm	Accuracy	Speed	Data Efficiency	P&ID Suitability	Adoption in P&ID
Two-stage (Faster R-CNN)	High	Low	Moderate	Good	Moderate
One-stage (YOLO, RetinaNet)	Moderate–High	High	Good	Good	High
Transformer (DETR, Def-DETR)	High	Moderate	Low	High potential	Emerging
Segmentation (Mask R-CNN, U-Net)	High	Low	Low	Complementary	Low

Despite the significant progress enabled by deep learning, several open challenges persist specifically within the P&ID domain. The scarcity of large-scale, publicly available annotated datasets constrains model training and fair benchmarking. Within-group fine-grained classification — the dominant residual error mode for visually similar symbol families — is not isolated by any of the reported evaluations, and is not architecturally addressed by any of the surveyed detector families. Furthermore, the detection of densely packed and heavily overlapping symbols in high-resolution diagram images continues to challenge even state-of-the-art detectors. These gaps motivate the methodology proposed in the following chapter.

3.4 End-to-End Digitization vs. Symbol Detection and Classification

A substantial fraction of the recent P&ID literature pursues *end-to-end digitization* rather than symbol detection and classification in isolation. Because both research lines share a common starting point (a raster P&ID image) and overlap in vocabulary, benchmarks, and citation patterns, the distinction between them is easy to miss — and missing it makes the literature appear more uniform, and the gaps less visible, than they actually are.

What end-to-end digitization works do. End-to-end digitization pipelines treat symbol detection as one module among several. The typical pipeline comprises: (i) a symbol detector that produces bounding boxes and class labels for each symbol instance; (ii) a line / pipeline tracer that vectorizes the pipeline network from raster pixels; (iii) a text recognition and association module that links alphanumeric tags to symbols; and (iv) a graph extraction step that infers edges between symbols based on pipeline connectivity. The output is a graph $G = (V, E)$ in which nodes V are symbols and edges E are pipeline connections, optionally annotated with tag information. The reported metrics typically include both symbol-level scores (precision, recall, mAP) and graph-level scores (edge accuracy, node–edge consistency).

Why this differs from the present study. Three consequences follow from the end-to-end framing, each of which separates such works from the scope of this study:

- **Aggregate symbol-detection evaluation:** The symbol-detection component of an end-to-end pipeline is evaluated only in aggregate (mAP across all classes), and the reported numbers reflect joint optimization with downstream modules. Within-group fine-grained classification quality is not isolated, and the dominant residual error mode — confusion among visually similar valve and instrument-bubble classes — is not separately measured or optimized for.
- **Vocabulary shaped by downstream consumers:** The symbol vocabulary used by end-to-end works is often constrained by what the downstream graph extraction module can consume. Coarse classes that produce unambiguous graph nodes are preferred; fine within-group distinctions that do not change the graph topology are often collapsed or dropped, biasing the evaluation toward coarse classification and hiding the fine-grained classification problem altogether.

- **Different technical focus:** The technical novelty of end-to-end works centers on line tracing, graph extraction, and structured prediction (e.g., the Relationformer architecture). These contributions are unrelated to fine-grained classification: they assume the symbol-detection module already provides correct fine labels and they do not improve the within-group decision boundary.

3.5 Literature Summary and Research Gaps

3.5.1 Literature Overview

Table 5 provides a structured summary of representative works in P&ID symbol detection and classification. Each row is tagged by *research line* — either **E2E digitization**, for works whose primary contribution is end-to-end graph extraction, or **Symbol det./class.**, for works whose primary contribution is symbol detection and classification itself. This tagging applies the distinction from Section 3.4: for rows tagged *E2E digitization*, only the symbol-detection component is summarized in the remaining columns; the corresponding graph-extraction, line-tracing, and text-linking contributions are noted in the original publications but are not benchmarked against in this study. The *Key limitation* column is correspondingly restricted to symbol-detection-and-classification limitations of each work.

Table 5: Summary of representative works in P&ID symbol detection and classification, tagged by research line. For works tagged E2E digitization, only the symbol-detection component is summarized.

	Cha et al. [39]	Paliwal et al. [4]	Paliwal et al. [5]	Kim et al. [7]	Stürmer et al. [6]
Approach	Symbol det./class.	E2E digitization	E2E digitization	E2E digitization	E2E digitization
Method	CNN classification	Kernel-based lines + fine-grained CNN	One-shot learning + GCN (OSSR-PID)	YOLOv5 + SE attention + SAHI	Relationformer (Deformable DETR)
Dataset	Private (real-world)	Dataset-P&ID (500 synthetic)	100 synthetic P&IDs	Private (real-world)	PID2Graph (real-world)
Cls.	9	32	32	30+	–
Performance	92.85% accuracy	>90% P/R	Comparable to supervised	High P/R on private data	Detection + edge metrics
Key limitation	Very narrow symbol set; classification only (no detection)	Synthetic only; aggregate metrics only; no within-group breakdown	Requires pre-segmented regions; aggregate metrics only	Not publicly reproducible; aggregate metrics only	Small dataset; symbol-detection component not isolated in evaluation

3.5.2 Research Gaps

A critical analysis of the surveyed literature reveals five significant research gaps that remain unaddressed by existing work:

Gap 1 — Absence of a standardized public benchmark: No large-scale, publicly available P&ID dataset with diverse real-world diagrams, comprehensive symbol vocabulary, and standardized evaluation protocols currently exists. The fragmented landscape of small and mostly synthetic datasets prevents meaningful cross-study comparison and slows community progress. Establishing a standard benchmark is a prerequisite for reproducible research in this domain.

Gap 2 — Unsolved cross-standard generalization: Virtually all existing models are trained and evaluated on diagrams matching a single symbol convention. The capacity to recognize symbols across different company standards, regional practices, or project eras — without retraining from scratch — has not been demonstrated. Domain adaptation and few-shot generalization across symbol vocabularies remain open problems.

Gap 3 — Underexploration of Transformer-based architectures: Despite the structural advantages of attention-based models for context-aware detection in dense, structured diagrams, Transformer-based detectors such as Deformable DETR and RT-DETR have received minimal investigation in the

P&ID domain. The potential of global self-attention for disambiguating visually similar symbols through contextual reasoning has not been systematically evaluated.

Gap 4 — Limited handling of dense and overlapping symbols: Existing detection pipelines — particularly those based on standard YOLO configurations — have not been systematically adapted to the high symbol density and frequent overlap characteristic of industrial P&ID sheets. Specialized strategies such as sliced inference [40], attention-guided feature refinement, or density-aware training remain underexplored in this context.

Gap 5 — Aggregate metrics hide fine-grained classification failure: The reported mAP and precision/recall scores in the P&ID literature do not isolate fine-grained classification quality on visually similar groups (the valve family, instrument-bubble family, fitting family). Within-group confusion — which is practically the dominant residual error mode on visually similar classes, and which carries the highest operational cost for engineering tasks — is therefore not directly measurable from the published numbers. As a consequence, the field lacks a shared protocol for diagnosing and comparing the fine-grained classification capability of detection systems, and no published P&ID work reports per-family accuracy, within-group confusion matrices, or rare-class macro-F1 alongside its aggregate detection scores.

Among the five gaps identified above, this study addresses Gaps 4 and 5 as its primary focus. Gap 4 (dense, overlapping symbols) is handled through overlap-preserving sliced inference, and Gap 5 (aggregate metrics conceal fine-grained classification failure) is addressed by introducing an FGC-specific evaluation protocol that reports within-group accuracy alongside the standard detection metrics. The study also contributes partial evidence toward Gap 1 by measuring synthetic-to-real generalization on a held-out real-world set. Gaps 2 and 3 — cross-standard generalization and Transformer-based detection — are acknowledged but left to future work. The following chapter presents the proposed methodology in detail.

4 Summary and Proposed Solutions

This chapter summarizes the existing problems in P&ID symbol detection and fine-grained classification, outlines the proposed approach this work takes to address them, and states the contributions the thesis aims to deliver.

4.1 Summary of Existing Problems

Through the problem analysis in Chapter 2.2, the survey of existing approaches in Chapter 3, and the gap analysis in Section 3.5.2, this work identifies the following core problems that the proposed approach must address:

- **Within-group fine-grained classification is the main source of remaining errors:** The P&ID symbol vocabulary contains visually similar groups — the valve family (gate, globe, ball, butterfly, needle, plug, pinch, diaphragm), the instrument-bubble family, the fitting family — whose members share a common base geometry and differ only in small internal markings. Standard one-stage detectors couple coarse super-category and fine within-group classification into a single softmax over the full vocabulary, devoting only a fraction of head capacity to the within-group decision, which in practice dominates the residual error rate on visually similar classes.
- **Aggregate metrics conceal the failure:** The detection metrics standardly reported in the P&ID literature (precision, recall, mAP) aggregate over all classes and therefore cannot distinguish a model that confuses random pairs of classes uniformly from a model whose errors are concentrated entirely within a single visually similar group. The dominant residual error mode (within-group confusion) is consequently hidden from current benchmarking, and no surveyed work isolates within-group accuracy or per-family classification quality (Gap 5).
- **Dense, high-resolution diagrams stress standard detection pipelines:** P&ID sheets routinely exceed $10,000 \times 7,000$ pixels and contain several hundred symbols per sheet, with frequent overlap and intersection with pipeline lines and annotation text. Standard detectors with fixed input resolution cannot ingest such diagrams without tiling, and naive tiling truncates symbols at patch boundaries, degrading recall (Gap 4).
- **Long-tail distribution starves rare fine classes at training time:** The fine-class frequency distribution is severely long-tailed. Standard cross-entropy training at full-diagram scale is dominated by gradient contributions from common classes, producing consistent under-fitting on rare fine classes — which are frequently the most safety-critical. Per-class data augmentation is impractical at full-image scale.
- **Data scarcity, especially real-world data:** Public P&ID datasets are small, mostly synthetic, and narrow in vocabulary. Real-world data is industrially confidential. Models trained exclusively on synthetic data face a synthetic-to-real domain gap that is rarely measured in the published literature.

4.2 Proposed Solutions Outlined in the Thesis

The proposal addresses the problems above through four connected design choices: a pipeline structure that separates coarse detection from fine-grained classification; a high-resolution input handling strategy compatible with that structure; a fine-grained classification stage whose training operates on cropped symbol regions so that rare-class balancing becomes practical; and a multi-phase dataset strategy with a real-world held-out evaluation set. Each commitment is described as a methodological choice with its justification, deliberately stopping short of specific architectures, parameter values, or training schedules — those decisions are deferred to the implementation roadmap in Chapter 5.

4.2.1 Pipeline Structure: Detection-then-Fine-Grained Classification

The central architectural commitment of the proposal is that symbol detection and fine-grained classification are treated as *two separable stages* rather than as a single coupled task with one large softmax over the full vocabulary. The detection stage is responsible for localization and *coarse classification* into symbol super-categories such as valve, instrument bubble, pump, and fitting. The fine-grained classification stage is responsible for *within-group discrimination* on localized symbol regions.

By decoupling the two sub-problems, dedicated model capacity is allocated to each: the coarse stage handles the easier discrimination problem, where features are global and class boundaries well-separated, while the fine-grained stage handles the harder within-group problem, where features are local. The separation also makes the fine-grained classification contribution directly measurable as an ablation — the system can be evaluated with and without the fine-grained stage on the same inputs — which is what Gap 5’s evaluation protocol requires. A single coupled detector offers no such ablation interface.

The overall pipeline groups input preprocessing, detection (localization plus coarse classification), fine-grained classification, and structured output into stages that can be developed, evaluated, and replaced independently. The design philosophy throughout prefers established, reproducible techniques over experimental alternatives, and stops at the level of describing *what each stage is responsible for* rather than prescribing a specific algorithmic implementation.

4.2.2 Detection Stage and High-Resolution Input Handling

The detection stage adopts a one-stage detector from the YOLO family as the primary direction of investigation, with YOLOv8 [27] as the initial baseline. The choice of detector family follows from the comparative analysis in Section 3.3: one-stage detectors offer a favourable balance of accuracy and inference speed, anchor-free heads avoid domain-specific anchor tuning, and the family is natively compatible with the tiling strategy described next. Two-stage detectors (Faster R-CNN) and Transformer-based detectors (Deformable DETR) are acknowledged as valid alternatives but rejected for this study — Faster R-CNN on inference-cost grounds, Deformable DETR on data-scale grounds, both documented in Section 3.3.

A key design decision concerns the *label set* used to train the detector. Because the architectural commitment is to leave within-group discrimination to the fine-grained stage, the detector can be trained on coarse super-category labels — with fine classes collapsed into their parent super-categories — rather than on the full fine-grained label set. Coarse-label training concentrates fine-grained learning capacity in the dedicated stage and simplifies the detector’s objective; full-fine-label training is the alternative configuration. The empirical comparison between the two is one of the planned ablations.

The high-resolution input handling problem (P3) is addressed through overlap-preserving tiling. Full-resolution diagrams are decomposed into fixed-size patches with a configurable overlap margin, following the Slicing Aided Hyper Inference (SAHI) strategy [40], and per-patch detections are subsequently merged using Weighted Boxes Fusion [41] rather than standard NMS. The overlap margin ensures that symbols crossing patch boundaries remain fully visible in at least one patch, and WBF preserves localization precision when the same symbol is detected from multiple overlapping patches. Preprocessing additionally applies standard scan-correction steps (resolution normalization, contrast enhancement[42], skew correction) to standardize input quality before tiling.

4.2.3 Fine-Grained Classification Stage

This stage is structured as a dedicated classifier that operates on cropped symbol regions and produces a within-group fine-class assignment, applied selectively to detections that belong to the visually similar groups (valve, instrument bubble, fitting) where coarse-only classification is in practice insufficient. The exact selection policy — which detections are routed through the fine-grained stage and which are passed through unchanged — is treated as an implementation decision rather than an architectural commitment, and is specified in the implementation roadmap.

The training-side property of the fine-grained stage is the methodological commitment that matters most at proposal level: training data is *cropped symbol regions* rather than full diagrams. This unlocks ag-

gressive per-class balancing (a balanced batch can be constructed by sampling one patch per class, regardless of the class’s natural frequency at diagram scale) and targeted augmentation for under-represented classes — both of which are impractical at full-image scale. Patch-scale training is the mechanism by which long-tail rare-class problem is addressed, and it is one of the practical reasons a separable stage is preferred over an end-to-end fine-grained detector.

The choice of classifier architecture, the use of metric-learning[43] or few-shot training objectives for rare-class robustness, and the comparison against per-group expert classifiers are treated as implementation-level investigations rather than fixed commitments. End-to-end fine-grained detectors — in which a single network is trained jointly with a fine-grained loss and no separable classification stage — are considered but rejected as the primary direction: they make rare-class balancing harder, they make the fine-grained contribution non-separable in evaluation, and they offer no clear data-efficiency advantage given the long-tail distribution. An end-to-end fine-grained detector remains a useful comparison point and is included as such in the implementation roadmap.

4.2.4 Dataset Strategy and Evaluation Protocol

Dataset is addressed through a three-phase strategy.

- Phase 1: reuses public synthetic data (Dataset-P&ID [4], supplemented with community datasets [8]) to establish a reproducible training baseline.
- Phase 2: expands vocabulary and diversity through a programmatic synthetic-generation pipeline that emits both coarse and fine labels per symbol (supporting the coarse-vs-fine detector ablation from Section 4.2.2) with configurable degradation augmentation to bridge the synthetic-to-real gap; targeted synthesis of under-represented fine classes follows the rare-class data recommendation noted as future work by Kim et al.[7].
- Phase 3: collects and annotates a small set of real-world diagrams (estimated 20–50 sheets) to serve as a *held-out evaluation set* for explicitly measuring the synthetic-to-real generalization gap; this real-world set is not used for training.

The evaluation protocol reports two families of metrics, in line with the FGC-specific protocol established in Section 2.2.3. The first family comprises standard detection metrics (precision, recall, per-class AP, mAP) for localization and aggregate classification quality. The second family comprises fine-grained classification metrics — within-group confusion matrices for each visually similar group, per-family accuracy (valve, instrument bubble, fitting), top- k accuracy on the FGC head evaluated on ground-truth crops, and macro-averaged F1 on rare fine classes. Reporting both families together is the methodological commitment that puts Gap 5 into practice; it is the protocol by which the proposal’s central contribution will be measured.

4.3 Expected Contributions of the Thesis

The proposed approach is expected to deliver contributions across three complementary domains: *technical* contributions concerning the artefacts and designs produced; *experimental* contributions concerning what the work measures and what those measurements reveal; and *application* contributions concerning what the work makes possible for downstream engineering use.

- **Technical contributions:**

- **A modular detection-plus-fine-grained-classification pipeline for P&IDs:** The pipeline architecturally separates coarse detection (localization plus super-category) from fine-grained within-group classification, in contrast to the single-softmax design of standard one-stage detectors and the joint-optimization design of end-to-end digitization systems (Section 4.2.1).

- **A fine-grained classification module designed for P&ID visually similar groups:** The module operates on cropped symbol regions with confidence- and super-category-based triggering, and its training data is constructed at patch scale to enable rare-class balancing that is impractical at full-image scale (Section 4.2.3).
- **A fine-grained-classification-specific evaluation protocol:** The protocol consists of within-group confusion matrices, per-family accuracy, top- k FGC-head accuracy, and rare-class macro F1, reported alongside standard detection metrics, directly addressing Gap 5 (Section 3.5.2).
- **A dataset construction strategy:** The strategy combines existing public synthetic data, a programmatic synthetic-generation pipeline that emits both coarse and fine labels with degradation augmentation, and a small real-world held-out evaluation set (Section 4.2.4).
- **Experimental contributions:**
 - **An ablation study quantifying the marginal contribution of fine-grained classification:** The study compares the detector-only configuration against the full detection-plus-FGC pipeline, and the coarse-label-trained detector + FGC against the fine-label-trained detector without FGC.
 - **An explicit measurement of the synthetic-to-real generalization gap:** The metric-by-metric comparison between the synthetic test set and the real-world held-out evaluation set is reported on both detection and FGC-specific metrics, with per-class and per-family breakdowns identifying where the gap concentrates — contributing to Gap 1.
- **Application contributions:** providing a more accurate symbol-recognition solution for practical problems such as safety-instrument counting and symbol-vocabulary coverage audits, particularly on rare but safety-critical classes where existing approaches are weak.

5 Implementation Roadmap

This chapter presents the implementation roadmap for the proposed P&ID symbol recognition research, structured across 15 weeks from environment setup and data preparation through to final evaluation and report submission.

5.1 Implementation Phases

Phase 1: Environment Setup and Data Preparation (Weeks 1–3)

- **Weeks 1–2:** Download and preprocess existing public datasets [4, 8]. Unify annotation formats to YOLO bounding box standard. Conduct exploratory data analysis to characterize symbol size distributions and class frequency imbalance.
- **Week 3:** Configure the deep learning environment (PyTorch, Ultralytics YOLOv8, SAHI). Implement the synthetic P&ID generation pipeline and produce an initial synthetic corpus with automatic bounding box annotations.

Phase 2: Baseline Model Training and Evaluation (Weeks 4–8)

- **Weeks 4–6:** Fine-tune COCO pre-trained YOLOv8 on the assembled P&ID dataset. Apply data augmentation strategy (mosaic, geometric transforms, photometric distortions). Monitor training convergence and adjust hyperparameters.
- **Weeks 7–8:** Integrate SAHI sliced inference pipeline and tune patch size and overlap parameters on the validation set. Evaluate the baseline model on the synthetic test set using mAP_{50} and per-class AP, with explicit attention to rare-class recall and long-tail performance. Produce a failure mode analysis report identifying frequently confused symbol class pairs and under-served rare classes.

Phase 3: Fine-Grained Classification and Ablation Study (Weeks 9–12)

- **Weeks 9–10:** Implement and train the fine-grained classification module (ResNet-34 cascade) for the visually similar symbol groups identified in Phase 2. Integrate the WBF deduplication module and tune fusion parameters.
- **Weeks 11–12:** Conduct structured hyperparameter optimization over confidence thresholds and augmentation parameters. Run systematic ablation study to quantify the individual contribution of each pipeline module (preprocessing, SAHI, fine-grained classification, WBF) to overall mAP_{50} .

Phase 4: Evaluation and Documentation (Weeks 13–15)

- **Week 13:** Apply the final pipeline to the real-world held-out evaluation set to assess the synthetic-to-real generalization gap. Analyze and visualize detection outputs on representative real-world diagrams.
- **Weeks 14–15:** Compile complete research documentation. Finalize the report, verify citation accuracy, and prepare the presentation for submission.

5.2 Implementation Schedule Summary

Table 6: *Detailed implementation schedule*

Phase	Weeks	Key Activities
1	1 – 2	Dataset assembly, annotation unification, EDA
	3	Environment setup, synthetic data generation pipeline
2	4 – 6	YOLOv8 baseline training with data augmentation
	7 – 8	SAHI integration, baseline evaluation, failure mode analysis
3	9 – 10	Fine-grained classification module, WBF integration
	11 – 12	Hyperparameter optimization, ablation study
4	13	Evaluation, qualitative result analysis
	14 – 15	Report finalization and presentation preparation

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